Functional Universality of Critical SK Autocorrelations

Physical Ram Coordinates, Dual-Energy Traces,
and Domain-Aware Approximation Certificates

Dr. Muralidhara Subbarao
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Stony Brook University
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Abstract

The pinned manuscript claims a common random functional limit of stationary critical Sherrington–Kirkpatrick spin autocorrelations for Gaussian and Rademacher disorder. This report examines its resolvent-trace step and gives an exact Ram moment reformulation with the actual conditioned equilibrium measure, physical Gram matrix, and closed form domain. The infinite source family is not a vector in ordinary $L^2(\Pi;\ell^2)$; its correct home is the dual of the vector energy space. We prove a deterministic angular tail estimate under explicit measure/form hypotheses, exact conforming resolvent lower bounds and dual-residual error identities, a relative matrix-error enclosure, and positive-time heat-trace bounds. We also give a concrete parity-reduced moment system and its linear-coordinate lower bound. These local results do not prove the manuscript's microscopic correctors, effective coefficient, varying-space convergence, or disorder universality. One hundred twenty finite floating-point checks validate specified algebraic identities, not the asymptotic theorem. Private Ram sources were read locally; this report and its audit should be treated as confidential working material.

Status and pin. The public source is OpenAI, October 5, 2026 [1], at commit:

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`

The assignment identifiers above are catalogue metadata, not a mathematical rating established here. The target is `07-trace.tex`, especially `eq:source-40`, `prop:trace-goe`, and `lem:trace-fixed`. Every substantive section of the build was read, including the support appendix. All 14 build files match the recorded SHA-256 and Git blob hashes. The local source directory is an archive, not a Git checkout.

Four kinds of assertion. **Source claims** are attributed to their source labels and are not certified wholesale by this report. **Conditional transfers** state the precise imported hypotheses. **Local results** have proofs below and are valid under their displayed assumptions. **Finite tests** are explicitly limited to small deterministic computations. No claim of priority, a new solution of critical SK dynamics, or a universal Ram convergence theorem is made.

Contents

1 The Pinned Problem and Its Normalizations **3**
1.1 Finite model, measure, and clock . . . 3
1.2 Basis-independent trace and rate conversion . . . 3

2 The Physical Limiting Measure and Domain **4**
2.1 Conditioning is not a product replacement . . . 4
2.2 Closed energy, operator, and the missing radial direction . . . 4
2.3 Effective coefficient and the target . . . 5

3 Stepwise Dependency Analysis of the Source **5**
3.1 From equilibrium to a closed dynamical form . . . 5
3.2 The target trace step . . . 6
3.3 Comparison, functional convergence, and support . . . 7

4 Direct Private Ram Correspondence **7**

5 Exact Ram Moment Systems for the Target **8**
5.1 A concrete cylinder bank and its physical matrices . . . 8
5.2 A precise finite-cylinder Green localization . . . 9
5.3 Full coordinates and a domain-aware vector formula . . . 9
5.4 Parity and a useful explicit lower bound . . . 10

6 Local Proof: The Infinite Source Lives in Dual Energy **10**
6.1 A deterministic angular tail certificate . . . 11

7 Local Proof: Compression and Error Certificates **12**
7.1 Exact lower bounds and residuals . . . 12
7.2 Relative matrix and load errors . . . 13
7.3 Why fixed-coordinate convergence is insufficient . . . 13

8 From Trace Control to Positive-Time Functions **14**
8.1 Explicit heat and derivative bounds . . . 14
8.2 A quantitative curve certificate from repeated resolvents . . . 14
8.3 A conditional end-to-end transfer . . . 16

9 Executed Finite Checks and Their Scope **16**

10 Public Literature, Benefits, and Limits **17**
10.1 Primary literature checked on October 7, 2026 . . . 17
10.2 What the Ram formulation usefully adds . . . 18
10.3 Costs, risks, and concrete next work . . . 18

11 Provenance and Reproduction **19**
11.1 Read scope and immutability . . . 19
11.2 Build and tool limitations . . . 19

1 The Pinned Problem and Its Normalizations

1.1 Finite model, measure, and clock

For $n \geq 2$, let $J_{ij} = J_{ji}$, $J_{ii} = 0$, with independent upper entries either $N(0,1)$ or uniform on $\{-1,1\}$. At zero field and inverse temperature $\beta = 1$, the probability measure on $\{-1,1\}^n$ is

$$\mu_{n,J}(\sigma) = Z^{-1} \exp\left(n^{-1/2} \sum_{i<j} J_{ij}\sigma_i\sigma_j \right). \tag{1.1}$$

Here Z is the sum normalizer. If uniform cube probability is the reference measure instead, its normalizer differs by 2^{-n}, but the probability measure is identical. Every site has an independent rate-one resampling clock. Set

$$h_i = n^{-1/2} \sum_{j \neq i} J_{ij}\sigma_j, \quad t_i = \tanh h_i, \quad w_i = 1 - t_i^2, \quad D_i f = \frac{f(\sigma^{i,+}) - f(\sigma^{i,-})}{2}.$$

If E_i is conditional expectation over spin i, then

$$H_n = \sum_i (I - E_i) = \sum_i (\sigma_i - t_i) D_i, \qquad \mathcal{E}_n(f,g) = \mu \sum_i w_i D_i f D_i g = \langle f, H_n g \rangle_\mu. \tag{1.2}$$

Indeed, condition on all spins except i and use their two-point variance. Thus $H_n \geq 0$ is self-adjoint in the physical space $L^2(\mu)$ and $P_t = e^{-tH_n}$. The flip rate is $(1 - \sigma_i t_i)/2$; a resampling ring need not flip the spin. Replacing that rate by one would change the model.

Define $T_n = n^{2/3}$ and

$$C_{n,J}(t) = n^{-1} \sum_i \langle \sigma_i, P_t \sigma_i \rangle_\mu, \qquad A_{n,J}(s) = n^{1/3} C_{n,J}(T_n s), \quad s > 0. \tag{1.3}$$

The randomness here is only disorder: equilibrium spins and the stationary dynamics have already been averaged. The source claims convergence in law in $C_{\mathrm{loc}}((0,\infty))$ under each entry law, to the same non-point-mass law of nonzero, nonnegative, nonincreasing functions tending to zero at infinity. Its further assertion is essential unboundedness of the limiting value at each fixed positive time. This is not convergence in probability to a deterministic curve, convergence including time zero, or an asserted uniform exponential decay rate. In fact $A_{n,J}(0) = n^{1/3}$.

1.2 Basis-independent trace and rate conversion

For Gaussian disorder only, adjoining independent $N(0, 2/n)$ diagonal entries to the matrix with off-diagonal entries $J_{ij}/\sqrt{n}$ gives a GOE matrix $W = U\Lambda U^t$. Its diagonal energy is constant on the cube; the molecular fields and H_n still omit this diagonal. Put

$$\lambda_1 \geq \lambda_2 \geq \cdots, \quad g_{a,n} = n^{2/3}(2 - \lambda_a), \quad x_{a,n} = n^{-1/3}(U^t\sigma)_a.$$

For any orthogonal U, not necessarily an eigenframe, orthogonal contraction of the feature index gives

$$B_n(z) = \sum_{a=1}^{n} \langle x_{a,n}, (z + T_n H_n)^{-1} x_{a,n} \rangle_\mu = \int_0^\infty e^{-zs} A_{n,J}(s)\, ds, \quad z > 0, \tag{1.4}$$

$$\sum_a \|x_{a,n}\|_\mu^2 = n^{1/3}, \qquad \sup_\sigma \left(\sum_a x_{a,n}(\sigma)^2 \right)^{1/2} = n^{1/6}. \tag{1.5}$$

The first identity follows from the spectral theorem for a nonnegative finite matrix, with the Gibbs inner product retained. It explains why a trace of observables, not the bare operator trace of $(z + T_n H_n)^{-1}$, is the relevant object.

If every site clock is multiplied by $r > 0$, then $H_n^{(r)} = rH_n$ and

$$A_n^{(r)}(s) = A_n^{(1)}(rs), \qquad B_n^{(r)}(z) = r^{-1} B_n^{(1)}(z/r). \tag{1.6}$$

A single rate-one global clock choosing a uniform site has operator H_n/n. Its matching critical physical time is $sn^{5/3}$, not $sn^{2/3}$. Discrete random-site updates are a separate discrete-time chain; the preceding identity is exact for its Poissonized version.

2 The Physical Limiting Measure and Domain

2.1 Conditioning is not a product replacement

The source defines increasing GOE edge points g_a as limits of $g_{a,n}$, fixing their scale and sign. On an admissible realization choose b with $b + g_1 > 0$ and set $l_a = (g_a + b)^{-1}$. Its spectral inputs imply

$$\sum_a l_a = \infty, \qquad \sum_a l_a^2 < \infty, \qquad g_a \geq c_g a^{2/3} \quad \text{for all sufficiently large } a. \tag{2.1}$$

For independent $Z_a \sim N(0, l_a)$, $S = \sum_a (Z_a^2 - l_a)$ converges almost surely and in L^2. The renormalized level is

$$D_g(b) = \lim_{L \to \infty} \left\{ \sqrt{L} - \sum_{a \geq 1} \left((g_a + b)^{-1} - (g_a + L)^{-1} \right) \right\}. \tag{2.2}$$

The source constructs this initially as a limit in probability in the edge law, with an almost-sure version along a deterministic subsequence. Conditioning on the zero-probability event $S = D_g(b)$ is defined through continuous densities, not by an unspecified conditional event. If q is the density of S and q_m that of its tail after m, then the first m marginal of Π_g has strictly positive density

$$\rho_m(y) = \frac{q_m(D_g(b) - \sum_{i \leq m}(y_i^2 - l_i))}{q(D_g(b))} \prod_{i \leq m} \frac{e^{-y_i^2/(2l_i)}}{\sqrt{2\pi l_i}}. \tag{2.3}$$

The consistency and b-independence of these marginals, the constraint $\sum_a (x_a^2 - l_a) = D_g(b)$, and their moment bounds are assertions proved in the source's `prop:edge-conditioned-law`. They are adopted as explicit inputs here. In particular, for every fixed integer $k \geq 0$,

$$\Pi_g |x_a|^k \leq C_{g,k} l_a^{k/2}, \tag{2.4}$$

with constants uniform in the retained coordinate. The parameter b is an auxiliary regularization of the conditioning construction, not a mass term to add to the heat-bath generator. Replacing Π_g by the independent Gaussian product changes the Gram matrices and the limiting dynamics.

2.2 Closed energy, operator, and the missing radial direction

Let $\mathcal{H} = L^2(\Pi_g)$ over the reals. Start with smooth compactly supported cylinder functions and the gradient form

$$\mathcal{E}_\Pi(F, G) = \int \sum_a \partial_a F \, \partial_a G \, d\Pi_g.$$

The source's `prop:gradient-domain` asserts closability and identifies the closure $V = D(\mathcal{E}_\Pi)$ through angular weak identities. For every finitely supported skew matrix A and admissible cylinder test ψ,

$$\int \psi(Ax)\cdot\nabla F\, d\Pi_g = -\int F\left[(Ax)\cdot\nabla\psi - \psi\sum_a (Ax)_a(g_a+b)x_a\right] d\Pi_g. \tag{2.5}$$

Finite rotations preserve the quadratic constraint, and the term with b cancels. This is not integration by parts against an infinite-dimensional Lebesgue density. The source proves $x \notin \ell^2$ almost surely using absolute continuity of a sufficiently remote tail with respect to a product Gaussian law. Two square-summable gradient vectors satisfying (2.5) differ by a multiple of x, hence agree. That observation explains uniqueness but not the whole cylinder-core theorem; the latter uses the source's compensated-derivative approximation argument.

The associated operator is defined by its domain:

$$D(H_\Pi) = \{u \in V : \exists h \in \mathcal{H}, \mathcal{E}_\Pi(u,v) = \langle h, v\rangle_\Pi \ \forall v \in V\}, \quad H_\Pi u = h. \tag{2.6}$$

The source proves $\ker H_\Pi$ consists of constants, $x_a \in V$, and $\nabla x_a = e_a$, so $\mathcal{E}_\Pi(x_a, x_b) = \delta_{ab}$. Membership of x_a in $D(H_\Pi)$ is not needed below and is not inferred from its form-domain membership. In particular, no formal expression $-\Delta + \nabla(-\log\Pi)\cdot\nabla$ is silently substituted for this closed operator.

2.3 Effective coefficient and the target

The microscopic energy measure in the source is the limit of $\omega_{a,n}(d\lambda) = \lambda\nu_{Y_a,H_n}(d\lambda)$ for $Y_a = (U^t\sigma)_a$. Its deterministic moments are prescribed by a conditioned-Haar experiment in `eq:source-3`. The source proves determinacy, a square-root exponential moment, and polynomial density in $L^2((1+\lambda)\omega + \delta_0)$. Its claimed time coefficient is

$$c_* = \omega(\{0\}) > 0, \qquad \omega([0,\infty)) = \mathbb{E}\,\mathrm{sech}^2(1+Z), \quad Z \sim N(0,1). \tag{2.7}$$

These two quantities must not be conflated. Finite systems have $\omega_{a,n}(\{0\}) = 0$, which does not forbid an atom in their weak limit. No numerical value of c_* is derived in this report.

With $L = c_* H_\Pi$, the target source formula and function are

$$B_g(z) = \sum_{a\ge 1}\langle x_a, (z+L)^{-1}x_a\rangle_\Pi, \qquad \mathcal{A}_g(s) = \sum_{a\ge 1}\langle x_a, e^{-sL}x_a\rangle_\Pi. \tag{2.8}$$

The terms are nonnegative, so finite partial sums have an unambiguous monotone limit; finiteness requires a theorem, not notation.

3 Stepwise Dependency Analysis of the Source

3.1 From equilibrium to a closed dynamical form

The following account uses actual source arguments rather than a routing list. The companion papers CS (*Critical slowing down in the Sherrington–Kirkpatrick model*, September 24, 2026) and CM (*Critical mixing in the Sherrington–Kirkpatrick model*, September 25, 2026) [2, 3] are not independently re-proved here. Their inputs are read as they are stated and used by M428.

1. **Section 02: edge law and cap inputs.** The GOE completion supplies a Haar eigenframe. Under a partition-function bias $L_\circ = Z_{\mathrm{cube}}/Z_{\mathrm{sp}}$, a rotated Gibbs spin has exactly the spherical law. The source uses an uncapped second-moment comparison to return to ordinary quenched

disorder. The cap $B_p = U \operatorname{diag}((\lambda_a - (2-p^2))_+)U^t$ and observation $h_p = B_p\sigma + N(0, B_p)$ have posterior interaction $W - B_p$. Companion CM supplies saddle, norm-density, field-curvature, and tail estimates; CS supplies edge resolvent estimates. The cap figure is schematic evidence of this definition, not numerical evidence of a law.

2. **Section 03: errors on observation paths.** The smallest scale is $r = n^{-1/3+\delta}$ with $\delta = 10^{-4}$. The source obtains an L^2 replica comparison, cube cylinder convergence, and $\mathbb{E}\|\nabla_{u_r} \log L_r\|^4 = o(1)$. Larger scales use the likelihood identity with drift one half of the squared score and bracket equal to its integral. Frobenius curvature error is integrated in $d \log p$. The unused frame is Haar only under the specified conditioning; intersecting a disorder event first could destroy that argument.

3. **Section 04: no power loss in variance.** The positive covariance error is compressed from parent to child cap. The radial deficit removes a rank-one obstruction; Haar compression gives an x^3 dimension factor against an x^{-2} inverse factor. Choosing the scale ratio small produces a contraction. The integrable correction $C(p^a + (np^3)^{-a})$ then yields, on one high-probability event for every test and every allowed scale,

$$Q_\mu \operatorname{Var}(f \mid h_p) \le Cp^{-2}\mathcal{E}_n(f) + Ce^{-n^\xi}\|f\|_\infty^2. \tag{3.1}$$

   The order "common event, then disorder-dependent optimizer" matters. A fixed error multiplying p^{-2} would instead cause a power loss.

4. **Section 05: a domain theorem, not a finite jet rule.** Angular weak identities are first closed and unique. The difficult reverse inclusion of that graph in the cylinder closure uses product charts, compensated derivatives, Hermite grades, and eventually constant coefficient arrays. The kernel argument uses tail triviality. Observation smoothings then have bounded slopes and converge weakly to this specific gradient.

5. **Section 06: relaxation and correctors.** Fixed-order connected Haar contractions give microscopic moments; a weighted difference recursion controls their growth. Polynomial Q_j with $Q_j(0) = 1$ approximate $\mathbf{1}_{\{0\}}$ in the weighted moment space. For a smooth compact cylinder F, the recovery functions are

$$f_{n,Q}(\sigma) = F(x_n) + n^{-1/3}\sum_a F_{,a}(x_n)(Q(H_n) - 1)Y_a. \tag{3.2}$$

   Their correction vanishes in L^2 but changes the scaled energy from the unrelaxed mass of ω to its atom. The lower bound in `eq:source-38` and recovery in `eq:source-39` use the same c_*. Polynomial degree is fixed before $n \to \infty$; exchanging these limits is not licensed. Positivity of c_* uses the earlier coarse gradient bound applied to the recovery functions, not an elementary Ram identity.

3.2 The target trace step

For a finite vector f, define $\|f\|_{z,n}^2 = z\mu|f|^2 + T_n\mathcal{E}_n(f)$, summing components. The resolvent variational identity is

$$\langle S, (z + T_nH_n)^{-1}S\rangle_\mu = \sup_f\{2\mu(S \cdot f) - \|f\|_{z,n}^2\}. \tag{3.3}$$

In `lem:trace-fixed`, a fixed source optimizer has bounded mass and scaled energy and satisfies $\|(z+T_nH_n)^{-1}x_{a,n}\|_\infty \le z^{-1}n^{1/6}$. The source applies the form lower bound to control the variational

upper limit and uses (3.2) for the reverse inequality. Source pairings require cylinder cutoff and moment convergence; weak convergence of coefficients alone is insufficient. This is a varying-Hilbert-space variational passage of the type studied by Kuwae–Shioya [9], restricted here to the admissible optimizer and recovery classes. The report does not upgrade it to an unrestricted Mosco theorem without checking that theorem's hypotheses.

For all coordinates simultaneously, Section 07 replaces the original pairing by its smallest-cap observation pairing. The geometric scale sum costs $O(n^{-\delta/2})$ in probability after the biased-to-quenched transfer. Spherical replacement has squared error of size $n^{\delta-1/20} = o(1)$. On the active cap let $R_0 = \sum_{j\le 50} u_j^2$. The source uses

$$\mathcal{V}_a = \sum_{j\le 50} \frac{u_j}{R_0 \delta_{aj}}(u_j\partial_a - u_a\partial_j), \qquad \delta_{aj} = n^{2/3} r^4 \frac{d_a - d_j}{B_{r,a} B_{r,j}},$$

where $d_a = 2 - \lambda_a$. Beyond a tight index cutoff, $\delta_{aj} \ge c a^{2/3}$. The three integration-by-parts terms differentiate the test, the vector-field coefficient, and $\log L_r$. They require, respectively, the smoothing gradient bound, inverse moments of R_0, and the fourth-moment score estimate. Their squared coefficients sum like $s_\ell = \sum_{a>\ell} a^{-4/3}$.

The source's quantitative conclusion, with deterministic constants only after high-probability localization, is

$$\limsup_{n\to\infty} \mathbb{P}\{B_{n,>\ell}(z) > \epsilon\} \le \eta + C_{\eta,z}\epsilon^{-1} s_\ell, \quad \ell \ge L_{\eta,z}, \quad \epsilon > 0. \tag{3.4}$$

One first takes $n \to \infty$, then $\ell \to \infty$, then $\eta \downarrow 0$. This is neither a deterministic error constant for an arbitrary disorder nor a disorder-averaged L^1 bound. The normalized tail optimizer is used only when $B_{n,>\ell}(z) \ge \epsilon$; by (1.5) its supremum bound is $n^{1/6}/(z\sqrt{\epsilon})$, making the uniform test estimates applicable. Our exact limiting-measure tail proof below does not replace this finite-n uniform estimate.

3.3 Comparison, functional convergence, and support

Section 07 also proves that the quadratic-feature trace with source $\sigma\sigma^t/n$ tends to zero for Gaussian disorder. The feature has pointwise Hilbert–Schmidt norm one. Section 08 first transfers this bounded trace through the iid Gaussian/sign mixture and only then transfers the spin trace. Matching the first three moments leaves fourth-order Taylor remainders; their averaged sensitivities involve $\sqrt{B_n^{[2]}(z)}$. A coordinate change or ordinary free-energy universality theorem does not control those derivatives of both the Gibbs measure and generator.

Section 09 divides the spectral measure by $1 + \lambda$. Joint transforms at positive z determine the resulting finite random measure on the one-point compactification of $[0, \infty)$. Heat tests then give C_{loc} convergence. Kernel constants and centered coordinates give large-time decay; finite shifts of the GOE edge law and exponential tilting of x_1^2 establish nondegeneracy. Section 10 constructs ordered finite edge shifts concentrating x_1 near ± 1 while suppressing other trace contributions, to place the constant-one function in the closed topological support. This does not contradict almost-sure decay, which is not a closed property in the positive-time local topology. None of these randomness/support claims follows from the Gram identity alone.

4 Direct Private Ram Correspondence

The following are actual local passages read, not inferred content of reports cited by those passages. They remain private references [4, 5, 6].

Private passage	Exact use in the present problem
Volume 21, `ch01d-physical-metrics.tex`, `v21c01:physical-gram-definition` and `v21c01:projected-matrix`	Synthesis Jc carries norm c^*Gc, $G = J^*J$. Here the physical measure is Π_g (or $\mu_{n,J}$), and stiffness is assembled from the closed energy. The coefficient generator is $G^{-1}K$ only with this metric.
Same chapter, `v21c01:residual-decomposition` and `v21c01:stability-residual`	A projected residual omits a perpendicular component. We use a form-dual residual, so atoms need only belong to V; the strong residual is used only if they belong to the operator domain.
Same chapter, `v21c01:approximate-gram` and finite measurements	Relative Loewner bounds are needed for quadrature/rounding. A finite positive Gram matrix is not an infinite-dimensional frame theorem.
Volume 21, `ch01f-spectral-limits.tex`, `v21c01:strong-not-norm`, `v21c01:stronger-contract`, `v21c01:dual-target`	Fixed-input strong resolvents do not imply uniform control on moving sources. Coercive form approximation does give our monotone resolvent lower bounds. The indefinite spectral-pollution example is not claimed to describe this positive shifted form.
TR8, Sections 1–4, especially `eq:operator-domain`, `eq:green-kernel`, `eq:duhamel-heat`	Define the inverse and its domain before any kernel. The present form domain is conditioned and infinite-dimensional, not a Euclidean Dirichlet domain. A heat semigroup need not have an assumed smooth kernel.
TR8, compatible panels, `eq:pullback`, `eq:weak-residual`, and discrete Green matrices	Moment reconstruction transforms mass, stiffness, and load together. For source coefficients the Green map has a mass factor; inverse stiffness alone acts on loads. Finite tests give a lower, not upper, bound on the full dual residual norm.

TR8's localization identity changes variables from y to $a = x - y$ and retains the source measure. We give a finite-cylinder version below. There is no translation-invariant measure on the infinite conditioned state space being postulated. The source uses classical variational, spectral, and angular techniques; calling their correct coordinate transport a Ram representation is not a claim that those techniques originated here.

5 Exact Ram Moment Systems for the Target

5.1 A concrete cylinder bank and its physical matrices

Fix an admissible g, a finite coordinate count m, and linearly independent smooth compactly supported atoms $\phi_1, \dots, \phi_N$ on $\mathbb{R}^m$. One concrete choice is an independent subset of $\chi_R(y)y^\alpha$, with a specified smooth cutoff; refinements also vary the cutoff and polynomial degree. A general dense bank may be taken from compact smooth cylinder tests. Define $Jc = \sum_j c_j\phi_j$ and

$$\begin{aligned} G_{ij} &= \int_{\mathbb{R}^m} \phi_i(y)\phi_j(y)\rho_m(y)\,dy, \\ K_{ij} &= \int_{\mathbb{R}^m} \nabla\phi_i(y)\cdot\nabla\phi_j(y)\rho_m(y)\,dy, \\ b_{i,a} &= \Pi_g[\phi_i x_a], \qquad M_z = zG + c_*K. \end{aligned} \tag{5.1}$$

Thus $G > 0$ by independence and the positive density, $K \geq 0$, and $M_z > 0$ for $z > 0$. When $a \leq m$, $b_{i,a}$ is the integral of $\phi_i(y) y_a \rho_m(y)$; when $a > m$, it is zero by sign symmetry of coordinate a. That zero is a statement about this restricted bank's load, not a zero true tail trace.

For a chosen number ℓ of sources, let b have columns b_a, $1 \leq a \leq \ell$. The weak resolvent approximation is

$$M_z c_a = b_a, \qquad u_{a,N} = J c_a, \qquad B_{\ell,N}(z) = \operatorname{Tr}(b^t M_z^{-1} b). \tag{5.2}$$

This is an explicit correspondence with source `eq:source-40`: the source is x_a, the physical integration measure is (2.3), and the operator domain is replaced only by a conforming subspace of its form domain. The sum is over observable coordinates, not basis atoms.

To use actual weak Ram moments, set $m_i(F) = \Pi_g[\phi_i F]$. On this finite trial space $m = Gc$ and $c = G^{-1} m$. Substitution into every test and energy yields

$$G_R = G^{-1}, \quad K_R = G^{-1} K G^{-1}, \quad b_R = G^{-1} b, \qquad (z G_R + c_* K_R) m_a = (b_R)_a. \tag{5.3}$$

The physical moment norm is $m^t G^{-1} m$, not $m^t m$. More generally, nonsingular reconstruction $c = Rm$ gives $G_R = R^t G R$, $K_R = R^t K R$, and $b_R = R^t b$; the trace in (5.2) is invariant by multiplication of these matrices. This proves the claimed finite Ram equivalence, rather than merely renaming a matrix solve. A rectangular reconstruction selects a subspace; a singular reconstruction requires a quotient or independent atom set.

If a source already has trial coefficients a, its load is Ga and the source-to-solution coefficient map is $(zG + c_* K)^{-1} G$. It is not $(zG + c_* K)^{-1}$ on those source coefficients. In the original coordinates the compressed generator $G^{-1} K$ is G-self-adjoint; its whitened representative $G^{-1/2} K G^{-1/2}$ is symmetric.

5.2 A precise finite-cylinder Green localization

For this fixed trial space and a source $q \in L^2(\rho_m dy)$, the projected resolvent has the finite-rank kernel

$$G_{z,N}(x, y) = \phi(x)^t M_z^{-1} \phi(y), \qquad u_N(x) = \int G_{z,N}(x, y) q(y) \rho_m(y)\, dy. \tag{5.4}$$

Take $h'(x, y) = G_{z,N}(x, y) \rho_m(y)$ and $h(u, a) = h'(u + a, u)$ in the private TR8 convention. Then, exactly,

$$u_N(x) = \int_{\mathbb{R}^m} h(x - a, a) q(x - a)\, da. \tag{5.5}$$

This is the Ram localization of an explicitly defined projected inverse, including its probability density. It is not an exact kernel for the full H_Π, nor is the density factor optional. Although the bank depends on only m coordinates, the coefficients ρ_m retain the infinite tail through q_m. Computing q_m is a separate approximation problem; substituting an independent Gaussian density does not solve it.

5.3 Full coordinates and a domain-aware vector formula

Take a countable linearly independent cylinder bank whose span is a form core, and complete finite coefficient sequences in the physical norm $\|c\|_G = \|Jc\|_\Pi$. Dense synthesis extends to an isometry onto $\mathcal{H}$. Denote this coefficient Hilbert space by $\mathcal{H}_G$ and define

$$H_R = J^{-1} H_\Pi J, \quad D(H_R) = J^{-1} D(H_\Pi), \quad V_R = J^{-1} V, \quad a_a = J^{-1} x_a.$$

The exact infinite Ram formula is consequently

$$B_g(z) = \sum_a \langle a_a, (z + c_* H_R)^{-1} a_a \rangle_G. \tag{5.6}$$

The definition is by the transported closed form, not by a formal infinite matrix product $G^{-1}K$ on all sequences. A Riesz basis in ordinary ℓ^2 is not being assumed. Each a_a exists individually in $\mathcal{H}_G$; the family $(a_a)_a$ need not belong to $\ell^2(\mathcal{H}_G)$. The next section gives the exact sense in which its aggregate pairing exists.

5.4 Parity and a useful explicit lower bound

Every individual sign flip of a coordinate preserves (2.3) and the cylinder energy. By closure it commutes with the form resolvent. Thus the sector odd in x_a and even in every other coordinate is invariant, and the source x_a belongs to it. Distinct such sectors are orthogonal in both mass and energy. A bank for source a can therefore use only that parity: for example an even cutoff times monomials with odd exponent at a and even exponents elsewhere. This reduces an actual matrix system without assuming independent coordinates.

In the conforming space $\operatorname{span}\{x_1, \dots, x_\ell\}$, write $v_a = \Pi_g x_a^2 > 0$. Sign symmetry gives $G = \operatorname{diag}(v_a)$, the source domain theorem gives $K = I$, and $b = \operatorname{diag}(v_a)$. Therefore

$$B_g(z) \;\geq\; \sum_{a \leq \ell} \frac{v_a^2}{z v_a + c_*}. \tag{5.7}$$

This exact variational lower bound is proved formally in the next section. It generally is not an equality: coordinate functions need not be generator eigenfunctions. Since $v_a \leq C_g l_a$, its infinite right side is summable by $\sum l_a^2 < \infty$. It supplies a nontrivial diagnostic and starting bank, not a substitute for the microscopic value of c_*.

6 Local Proof: The Infinite Source Lives in Dual Energy

Fix $z, c > 0$ and a nonnegative closed form $\mathcal{E}$ on a Hilbert space $\mathcal{H}$, with operator H. Give $V = D(\mathcal{E})$ the Hilbert norm

$$\|u\|_{V_z}^2 = z\|u\|_{\mathcal{H}}^2 + c\mathcal{E}(u), \qquad \mathcal{V}_z = \ell^2(V_z).$$

All dual norms below refer to this shifted form norm, including c.

Theorem 6.1 (Dual-energy trace identity)**.** *Let $q_a \in \mathcal{H}$, with countably many indices. On finite vectors define $\Lambda(f) = \sum_a \langle q_a, f_a \rangle$. The following are equivalent:*

1. *Λ extends boundedly to $\mathcal{V}_z$;*
2. *$\sum_a \|q_a\|_{V_z'}^2 < \infty$;*
3. *$B(z) = \sum_a \langle q_a, (z + cH)^{-1} q_a \rangle < \infty$.*

In that case

$$B(z) = \|\Lambda\|_{\mathcal{V}_z'}^2 = \sup_{f \in \mathcal{V}_z} \{2\Lambda(f) - \|f\|_{\mathcal{V}_z}^2\} = \sum_a \|(z + cH)^{-1/2} q_a\|_{\mathcal{H}}^2. \tag{6.1}$$

The optimizer is $((z + cH)^{-1} q_a)_a \in \mathcal{V}_z$.

Proof. For one source q, the Riesz representative $u \in V_z$ of $v \mapsto \langle q, v\rangle$ solves $z\langle u, v\rangle + c\mathcal{E}(u, v) = \langle q, v\rangle$. The operator definition identifies $u = (z + cH)^{-1}q$; testing with u gives $\|q\|^2_{V_z'} = \|u\|^2_{V_z} = \langle q, u\rangle$. For finitely many sources the direct-sum Riesz theorem adds these squares. If the sums stay bounded, their representatives are Cauchy in $\mathcal{V}_z$, define a bounded functional there, and converge to the displayed optimizer. Conversely, restrictions of a bounded functional to each finite direct sum have uniformly bounded squared norms, so the series converges. Completing the square proves the supremum formula. The last equality follows from the nonnegative spectral functional calculus. No ordinary square-summability of the unregularized sources is used. □

For $q_a = x_a$ and $\mathcal{H} = L^2(\Pi_g)$, the source's $x \notin \ell^2$ almost surely implies $\sum_a \Pi_g x_a^2 = \infty$ by monotone convergence. Thus $x = (x_a)_a$ is *not* in $L^2(\Pi_g; \ell^2)$, whereas Theorem 6.1 will make it a dual-energy vector. In Ram coordinates the same statement holds in $\ell^2(V'_{R,z})$, with the physical G norm. A notation such as $\langle a, (z + cH_R)^{-1}a\rangle$ must mean this direct-sum dual pairing or its increasing finite restrictions, not an ordinary ℓ^2 inner product of a nonexistent vector.

6.1 A deterministic angular tail certificate

Here is a local proof on the limiting conditioned measure. Its hypotheses are named so that this is not mistaken for a proof of the whole SK limit.

Theorem 6.2 (Angular dual tail)**.** *Assume Π_g, V, and its cylinder core have the properties in Section 2, including (2.5). Fix $m \geq 9$ and a sufficiently large $\ell \geq m$. Set $R_0 = \sum_{j\leq m} x_j^2$ and, for $a > \ell$,*

$$\delta_{aj} = g_a - g_j > 0, \qquad \kappa_a = \max_{j\leq m} \delta_{aj}^{-1}, \qquad V_a = \sum_{j\leq m} \frac{x_j}{R_0\delta_{aj}}(x_j\partial_a - x_a\partial_j).$$

Assume $\Pi_g R_0^{-4} < \infty$ and $\sup_a \Pi_g x_a^4 < \infty$. Then, for every $z, c > 0$,

$$\sum_{a>\ell} \langle x_a, (z + cH_\Pi)^{-1}x_a\rangle_\Pi \leq \sum_{a>\ell} \left(\frac{\Pi_g|V_a|^2}{c} + \frac{\Pi_g|\operatorname{div} V_a|^2}{z} \right) \leq C_{g,m}(c^{-1} + z^{-1}) \sum_{a>\ell} \kappa_a^2. \tag{6.2}$$

With (2.1), this is at most $C_{g,m,z,c}\ell^{-1/3}$ after increasing the starting index. In particular, the full trace is finite at every positive z.

Proof. The field is tangent to the finite-coordinate spheres. Applying it to the Gaussian logarithmic density gives exactly $-x_a$, since each rotation contributes $-(g_a - g_j)x_a x_j$. The radial conditioning factor is unchanged by the rotations. Expanding its ordinary coordinate divergence gives

$$\operatorname{div} V_a = -x_a \left(\frac{\sum_{j\leq m} \delta_{aj}^{-1}}{R_0} - \frac{2\sum_{j\leq m} x_j^2\delta_{aj}^{-1}}{R_0^2} \right). \tag{6.3}$$

Consequently (2.5), with these coefficients inserted, implies

$$\Pi_g[x_a f] = \Pi_g[V_a f] + \Pi_g[f \operatorname{div} V_a]. \tag{6.4}$$

For rigor first multiply the coefficients by a cutoff $\chi(R_0/\varepsilon)$ vanishing near zero, and by a radial cutoff at infinity in a finite coordinate set containing the test coordinates and $1, \ldots, m, a$. The latter is annihilated by the rotations. The derivative of the former is bounded by $C\kappa_a|x_a|R_0^{-1}\mathbf{1}_{\{R_0\leq 2\varepsilon\}}$. Its L^2 norm tends to zero by the assumed inverse and fourth moments and dominated convergence. Ordinary moment bounds remove the outer cutoff. Thus (6.4) holds on the cylinder core.

Pointwise, the a component of V_a is $\sum_j x_j^2/(R_0\delta_{aj})$ and its j component is $-x_a x_j/(R_0\delta_{aj})$. Hence

$$|V_a|^2 \le \kappa_a^2(1+x_a^2/R_0), \qquad |\operatorname{div} V_a|^2 \le (m+2)^2\kappa_a^2 x_a^2/R_0^2.$$

Cauchy–Schwarz bounds their expectations by constants times κ_a^2. The right side of (6.4) is continuous in V_z, so the core extends that identity to V. For a finite index set $I \subset \{\ell+1,\dots\}$, apply Cauchy–Schwarz to the combined gradient and mass pairings in (6.4):

$$\left|\sum_{a\in I}\Pi_g[x_a f_a]\right|^2 \le \left[\sum_{a\in I}\left(c^{-1}\Pi_g|V_a|^2 + z^{-1}\Pi_g|\operatorname{div} V_a|^2\right)\right]\sum_{a\in I}(c\mathcal{E}_\Pi(f_a)+z\|f_a\|_\Pi^2).$$

Theorem 6.1 and monotone convergence over I prove the first inequality. For fixed m, $g_a - g_j \ge c_g' a^{2/3}$ eventually, and $\sum_{a>\ell} a^{-4/3} \le \int_\ell^\infty t^{-4/3}dt = 3\ell^{-1/3}$. Finitely many omitted sources each have finite resolvent norm at most $z^{-1}\Pi_g x_a^2$. □

The inverse-moment hypothesis is available under the source's measure construction: (2.3) is bounded by a constant times its finite Gaussian density, hence bounded near zero. Integration of $|y|^{-8}$ in $m > 8$ dimensions is finite, and outside the unit ball $R_0^{-4} \le 1$. The fourth-moment condition follows from (2.4) and $\sup l_a < \infty$. Taking $m = 50$ matches the source's convenient margin. The constants depend on the realized edge data and on density and inverse-moment bounds. No deterministic universal constant, integrability in disorder, or computed numerical value for those constants is claimed.

7 Local Proof: Compression and Error Certificates

7.1 Exact lower bounds and residuals

Let $q_a \in \mathcal{H}$, $1 \le a \le \ell$, and let $V_N \subset V$ be a finite conforming trial space with physical matrices as in (5.1). Write $a_z(u,v) = z\langle u,v\rangle + c\mathcal{E}(u,v)$.

Theorem 7.1 (Trace error is a dual residual)**.** *Let $u_a = (z+cH)^{-1}q_a$ and let $u_{a,N}$ be its exact Galerkin solution. Then*

$$\sum_{a\le\ell}\langle q_a,u_a\rangle - \operatorname{Tr}(b^t M_z^{-1}b) = \sum_{a\le\ell}\|u_a-u_{a,N}\|_{V_z}^2 = \sum_{a\le\ell}\|r_{a,N}\|_{V_z'}^2 \ge 0, \tag{7.1}$$

where $r_{a,N}(v) = \langle q_a,v\rangle - a_z(u_{a,N},v)$ for every $v \in V$. For nested spaces with union a form core, the trace approximations increase to the finite-source trace. If an infinite source tail is bounded by $\tau_\ell(z)$, then the full trace satisfies

$$B_{\ell,N}(z) \le B(z) \le B_{\ell,N}(z) + \sum_{a\le\ell}\|r_{a,N}\|_{V_z'}^2 + \tau_\ell(z). \tag{7.2}$$

Proof. Galerkin orthogonality says $a_z(u_a - u_{a,N}, v_N) = 0$ for every $v_N \in V_N$. The exact weak equation gives $\langle q_a,u_a\rangle = a_z(u_a,u_a)$ and the discrete one gives $\langle q_a,u_{a,N}\rangle = a_z(u_{a,N},u_{a,N}) = b_a^t M_z^{-1} b_a$. Pythagoras in the form inner product proves the first equality in (7.1). The Riesz representative of $r_{a,N}$ is $u_a - u_{a,N}$, proving the second. Orthogonal projection onto increasing closed subspaces yields monotonicity; core density makes the errors tend to zero for each fixed a. Finally split B into its first ℓ sources and its nonnegative tail. □

The weak residual requires no $Hu_{a,N}$ in $\mathcal{H}$. If that stronger domain condition does hold, put $d_a = q_a - (z + cH)u_{a,N} \in \mathcal{H}$. The spectral theorem yields the usable but often looser bound

$$\|r_{a,N}\|^2_{V_z'} = \langle d_a, (z + cH)^{-1} d_a \rangle \le z^{-1} \|d_a\|^2. \tag{7.3}$$

A residual tested only inside V_N is zero by construction; it is not the norm on the right of (7.2). Testing in a larger finite space gives a lower bound on the true dual norm, unless a complementary upper bound is also supplied. This is exactly the distinction made in the private physical-metric and weak-residual passages.

For approximate coefficients y_a, a safer algebraic lower functional is

$$\mathcal{L}(y) = \sum_{a \le \ell} (2b_a^t y_a - y_a^t M_z y_a).$$

Completing the square in the full form space proves $B_\ell - \mathcal{L}(y) = \sum_a \|\langle q_a, \cdot \rangle - a_z(Jy_a, \cdot)\|^2_{V_z'}$. Thus a solver residual, quadrature error, trial error, and source tail need not be mixed into one unexplained tolerance.

7.2 Relative matrix and load errors

Proposition 7.2 (A finite matrix enclosure)**.** *Suppose $M > 0$, $0 \le \varepsilon < 1$, and $(1-\varepsilon)M \preceq \widehat{M} \preceq (1+\varepsilon)M$. Let $b, \widehat{b}$ be load matrices and suppose $\|M^{-1/2}(b - \widehat{b})\|_{\mathrm{HS}} \le \eta$. For $B_N = \mathrm{Tr}(b^t M^{-1} b)$ and $\widehat{B} = \mathrm{Tr}(\widehat{b}^t \widehat{M}^{-1} \widehat{b})$,*

$$\left[\max\{0, \sqrt{(1-\varepsilon)\widehat{B}} - \eta\}\right]^2 \le B_N \le \left[\sqrt{(1+\varepsilon)\widehat{B}} + \eta\right]^2. \tag{7.4}$$

Proof. Inversion reverses Loewner order. Applying the resulting inequalities to each column of $\widehat{b}$ places $\|M^{-1/2}\widehat{b}\|^2_{\mathrm{HS}}$ between $(1-\varepsilon)\widehat{B}$ and $(1+\varepsilon)\widehat{B}$. The triangle and reverse triangle inequalities with the load error give (7.4). □

These are hypotheses to validate, not automatic outputs of quadrature. Absolute entrywise error in a nearly singular physical Gram direction does not imply the relative form bound. Combining (7.4), a full residual upper estimate, and Theorem 6.2 produces a genuine conditional trace enclosure. The present finite floating-point experiment does not furnish interval bounds for the infinite conditioned density or c_*.

7.3 Why fixed-coordinate convergence is insufficient

On $\mathcal{H} = \ell^2(\mathbb{N})$ let $H = I$ and use the finite-index sources $q_{a,n} = \mathbf{1}_{\{a=n\}} e_n$, $1 \le a \le n$. Every fixed-coordinate resolvent contribution tends to zero and the operator itself does not change, but the summed trace equals $(z+1)^{-1}$ for every n. The mass escapes in the observable index. Thus even exact strong resolvent convergence of operators cannot remove the tail requirement. This simple positive example is more directly relevant here than importing an indefinite spectral-pollution mechanism.

8 From Trace Control to Positive-Time Functions

8.1 Explicit heat and derivative bounds

Let $\nu = \sum_a \nu_{q_a,cH}$ be the nonnegative spectral measure of the source family, so $B(z) = \int (z+\lambda)^{-1} d\nu(\lambda) < \infty$. For $s \geq s_0 > 0$,

$$\sum_{a>\ell} \langle q_a, e^{-scH} q_a \rangle \leq C(s_0, z) B_{>\ell}(z), \qquad C(s_0, z) = \begin{cases} z, & s_0 z \geq 1, \\ e^{s_0 z - 1}/s_0, & s_0 z < 1. \end{cases} \tag{8.1}$$

Indeed maximize $(z+\lambda)e^{-s_0\lambda}$ over $\lambda \geq 0$; its maximum occurs at zero or at $s_0^{-1} - z$. For each integer $k \geq 0$,

$$|\mathcal{A}^{(k)}(s)| \leq \sup_{\lambda \geq 0} (z+\lambda)\lambda^k e^{-s_0\lambda}\, B(z), \qquad s \geq s_0. \tag{8.2}$$

Dominated differentiation proves smoothness and complete monotonicity: $(-1)^k \mathcal{A}^{(k)}(s) \geq 0$. In particular the angular tail bound also controls the infinite-coordinate heat tail uniformly away from zero. It does *not* say that the heat evolution of a Galerkin compression is a pointwise lower bound: exponential functional calculus does not follow from the resolvent variational inequality in that manner.

If $\nu(\{0\}) = 0$, fix $s_0 > 0$ and dominate $e^{-s\lambda}$ by $e^{-s_0\lambda}$ to obtain $\mathcal{A}(s) \to 0$ as $s \to \infty$. For Π_g, the imported kernel theorem and centered x_a give this zero-atom condition. No spectral gap is proved by this argument. For one coordinate with variance v_a and form energy one, its normalized spectral measure has mean c_*/v_a; scalar Jensen gives

$$\mathcal{A}_g(s) \geq v_a e^{-sc_*/v_a} > 0. \tag{8.3}$$

This checks positivity, not non-point-mass randomness. Randomness still requires information about the edge law, such as the source's finite-shift argument.

8.2 A quantitative curve certificate from repeated resolvents

The parent supplied a local proof and finite evidence for the following extension [15]. The proof was read and checked for this report; its checker was not rerun. This is a scalar spectral consequence, not another claimed SK limit theorem. We use p for its rational approximation order, distinct from the cylinder dimension used earlier.

Theorem 8.1 (Weighted backward-Euler trace bound)**.** *For any nonnegative measure ν on $[0,\infty)$ with $B(z) = \int (z+\lambda)^{-1} d\nu < \infty$, let*

$$A(t) = \int e^{-t\lambda} d\nu, \qquad E_p(t) = \int (1 + t\lambda/p)^{-p} d\nu, \quad p \in \mathbb{N},\ p \geq 2.$$

For every $a > 0$ and all $t \geq a$,

$$0 \leq E_p(t) - A(t) \leq \frac{K_E(a,z)}{p} B(z), \qquad K_E(a,z) = \frac{8z}{e^2} + \frac{108}{ae^3}. \tag{8.4}$$

Moreover $E_p(t)$ decreases to $A(t)$ as $p \to \infty$.

Proof. Put $u = t\lambda$ and $v = p\log(1+u/p)$. For $0 \leq u \leq p$, $u - u^2/(2p) \leq v \leq u$ and $v \geq u/2$. Therefore

$$0 \leq e^{-v} - e^{-u} \leq (u-v)e^{-v} \leq \frac{u^2}{2p} e^{-u/2}.$$

Multiplying by $z + u/t$ and using the maxima $\sup u^2 e^{-u/2} = 16/e^2$ and $\sup u^3 e^{-u/2} = 216/e^3$ gives $K_E(t,z)/p$. For $u \geq p$, the derivative of $(z+u/t)(1+u/p)^{-p}$ has the sign of $1 - tz - u(1-1/p) \leq 0$. Its value is at most $(z + p/t)2^{-p}$. The integer inequalities $p2^{-p} \leq 1/2$ and $p^2 2^{-p} \leq 9/8$ for $p \geq 2$ follow by checking $p = 2, 3$ and the successive ratios thereafter. Thus this second bound is at most $(z/2 + 9/(8t))/p \leq K_E(t,z)/p$. Integration against $(z+\lambda)^{-1} d\nu$ proves the bound and finiteness of E_p. Finally $p \log(1 + u/p)$ increases with p, since its derivative is $\log(1+x) - x/(1+x) \geq 0$, $x = u/p$. The error bound or dominated convergence gives the limit. □

Let ν_ℓ be the first ℓ source measures. Suppose $B_\ell(z) \leq B_0$, $B_{>\ell}(z) \leq \epsilon$, and a computed curve satisfies $|\widehat{E}(t) - E_{p,\ell}(t)| \leq \delta$ on a specified time set in $[a, \infty)$. Nonnegativity and (8.1) give the bracket

$$\begin{aligned} \max\{0, \widehat{E}(t) - \delta - K_E(a,z)B_0/p\} &\leq \mathcal{A}(t) \\ &\leq \widehat{E}(t) + \delta + C(a,z)\epsilon. \end{aligned} \tag{8.5}$$

The lower bound uses $\mathcal{A} \geq \mathcal{A}_\ell$ and the rational error; the upper bound uses $\mathcal{A}_\ell \leq E_{p,\ell}$ and the source tail. The same statement holds pointwise if only pointwise numerical error bounds are available. An upper bound B_0 requires a certificate such as (7.2); the lower Galerkin trace alone is not an admissible B_0.

For even $p = 2k$, let $R_h = (I + hL)^{-1}$ with $h = t/p$ and $L = c_* H_\Pi$. Then

$$E_{p,\ell}(t) = \sum_{a \leq \ell} \|R_h^k x_a\|_\Pi^2.$$

To certify a numerical repeated solve in the full physical space, start $\widehat{u}_0 = x_a$, take $\widehat{u}_j \in D(L)$, and compute full residuals $r_j = (I + hL)\widehat{u}_j - \widehat{u}_{j-1}$. Because $\|R_h\| \leq 1$, induction gives

$$\begin{aligned} \|\widehat{u}_k - R_h^k x_a\| &\leq \eta_{a,k} := \sum_{j=1}^{k} \|r_j\|, \\ \left| \|\widehat{u}_k\|^2 - \|R_h^k x_a\|^2 \right| &\leq \eta_{a,k}(2\|\widehat{u}_k\| + \eta_{a,k}). \end{aligned} \tag{8.6}$$

Indeed the error recurrence is $e_j = R_h e_{j-1} + R_h r_j$; the squared norm estimate is the triangle inequality followed by factoring a difference of squares. Summing over the finite source set gives δ. Form-only trial functions require a corresponding coercive form-dual residual norm and a full complementary-space majorant, not the undefined strong residual. Uniformity over a time interval must be proved for the residual bounds, or via controlled interpolation; sample times alone do not establish it.

Finally the source's probabilistic trace-tail estimate has a direct positive-time consequence. Here $A_{n,>\ell}(s) = \sum_{j>\ell} \langle x_{j,n}, e^{-sT_n H_n} x_{j,n} \rangle_\mu$. For every $a, z, \epsilon > 0$, for sufficiently large ℓ after the same localization,

$$\limsup_n \mathbb{P}\left\{ \sup_{s \geq a} A_{n,>\ell}(s) > \epsilon \right\} \leq \eta + C_{\eta,z} C(a,z) \epsilon^{-1} \sum_{j>\ell} j^{-4/3}. \tag{8.7}$$

This follows by applying (8.1) before (3.4); the finite edge decomposition here is Gaussian. Derivative tails use the analogous multiplier in (8.2). No deterministic quenched rate follows without deterministic constants. Powers of a projected matrix need their own residual analysis for δ; there is still no automatic Galerkin heat monotonicity.

8.3 A conditional end-to-end transfer

For clarity, the precise logical contract for recovering the source result from a Ram computation is as follows.

1. Supply the random conditioned laws and their static moment convergence, the closed cylinder form and its domain/core, and the deterministic coefficient $c_* > 0$.
2. Prove the finite-model lower/recovery bounds for every optimizer and test actually used, with a common quenched event and polynomial size control. This gives joint convergence of each finite-source trace.
3. Supply (3.4). Then finite-source convergence and tail truncation give joint convergence of the full Gaussian traces at each finite set of positive z. Limiting finiteness can also be obtained locally by Theorem 6.2 under its hypotheses.
4. Supply the quadratic-feature vanishing and fourth-derivative comparison through the complete iid mixture; this transfers the full trace laws to Rademacher disorder without a new eigenframe assumption for that model.
5. Use joint transforms, not only their means, to identify the random divided spectral measures and then the positive-time functions.

For the last step set $d\eta_n = (1+\lambda)^{-1} d\nu_n$ on $K = [0, \infty]$. The mass is $B_n(1)$, hence tight. The functions $k_z(\lambda) = (1+\lambda)/(z+\lambda)$, extended by $k_z(\infty) = 1$, form a determining family: their closed linear span contains constants and resolvents, whose products are obtained by partial fractions and uniform limits. Stone–Weierstrass gives density in $C(K)$. Joint convergence at a countable dense set of z therefore identifies every subsequential random-measure law, including exclusion of mass at infinity when the limiting transform is that of ν_g. The heat tests $(1+\lambda)e^{-s\lambda}$ vanish at infinity and form a uniformly continuous family in sup norm for s in a compact positive interval. A finite net of these tests gives uniform convergence there. This proves the stated conditional transfer to C_{loc}.

An exact Ram identity establishes none of the probabilistic inputs in items 1–4. The order of limits in a numerical program must also separate system size n, source cutoff ℓ, cylinder dimension m, trial dimension N, polynomial-corrector index, quadrature precision, and time endpoint s_0. A fixed finite collection of solved systems does not exchange these limits.

9 Executed Finite Checks and Their Scope

The owned script `_audit/M428-author/checks.py` uses the bundled Python runtime, NumPy double precision, and seed 428. It enumerates all 2^n spin states for $n = 4, 5, 6$, for one generated coupling matrix of each Gaussian and sign law at each size. The Gibbs probabilities are evaluated by normalized exponentials. Both the direct heat-bath matrix and the independently assembled half-difference Dirichlet matrix are formed. The full nonorthogonal test bank consists of Walsh monomials followed by a nonsingular triangular mixing. Reduced spaces retain the constant and odd monomials of degree at most 1, 3, or 5. All matrices use their actual Gibbs weights. No continuum conditioned measure is sampled.

n	Law	$B_n(0.7)$	Degree-1 gap	Degree-3 gap	Wrong-mass defect
4	Gaussian	0.90249311	$1.3304\,10^{-3}$	$< 10^{-14}$	0.25359
4	Sign	0.93328983	$1.4543\,10^{-3}$	$< 10^{-14}$	0.33498
5	Gaussian	1.14468610	$3.6143\,10^{-3}$	$< 10^{-14}$	0.77634
5	Sign	0.91994378	$1.6444\,10^{-3}$	$< 10^{-14}$	0.32780
6	Gaussian	0.81750650	$2.4422\,10^{-3}$	$4.0548\,10^{-7}$	0.21243
6	Sign	1.11484009	$7.5856\,10^{-3}$	$7.0303\,10^{-5}$	0.80110

Here "gap" is exact-matrix trace minus reduced trace in floating-point arithmetic, not a confidence interval. Tiny negative gaps of order 10^{-15} in fully resolved spaces are rounding. "Wrong-mass defect" is the discrepancy after deliberately replacing the full nonorthogonal mass matrix by the identity while keeping stiffness and loads unchanged. Its size concretely detects a false Euclidean interpretation of Ram coefficients.

All **120 assertions passed**. The tests cover detailed balance, constant annihilation, positivity, the two independent form assemblies, spectral-trace equality, total source mass $n^{1/3}$, orthogonal feature invariance, full nonorthogonal Gram invariance, nested variational lower bounds, the dual-residual identity, clock multiplication by 2.3, and the heat bound at $s_0 = 0.4$. An angular-field test checks tangency, the Gaussian log-density derivative, and the explicit divergence against a centered finite difference. Its tolerance is $2 \cdot 10^{-7}$; the other assertion tolerance is $2 \cdot 10^{-9}$.

A separate diagonal toy uses $He_j = j^{2/3}e_j$ and $q_j = j^{-1/3}e_j$. Its unregularized squared source norm diverges, while $\sum j^{-2/3}/(z + cj^{2/3})$ converges and has tail bounded by $3c^{-1}\ell^{-1/3}$. The integral comparison proves this statement; finite sums through 10,000 terms merely check its implementation at $\ell = 10, 100, 1000$. This toy is not Π_g and is not evidence for independent edge dynamics.

The separately supplied parent evidence reports **58,374 assertion groups passed**: 798 exact rational dyadic inequalities, 360 exact rational scalar residual-chain assertions, 28,416 floating scalar error tests, 28,416 floating monotonicity tests, and 384 finite atomic-measure brackets. The parent script and result JSON were inspected, their script hash is checked against the JSON, and their hashes are included as read-only inputs to this report's manifest. These counts are not added to our 120 author assertions and do not certify continuous parameter ranges by enumeration. The continuous claims follow from the displayed proofs.

The full JSON results retain all defects and tolerances. These experiments are neither interval-arithmetic certificates nor a statistical universality test. They do not estimate c_*, prove the GOE edge law, validate the companion cap inequalities, or establish any $n \to \infty$ rate.

10 Public Literature, Benefits, and Limits

10.1 Primary literature checked on October 7, 2026

A current public-topic search located and checked primary bibliographic records and abstracts, with explicit limits on reading depth recorded in the URL audit. Billoire–Campbell [7] is numerical equilibrium relaxation work, not a proof of the present random functional limit. Ramirez–Rider–Virag [8] supplies the stochastic Airy spectral context. Kuwae–Shioya [9] is the varying-space spectral-structure framework, not a source of model-specific correctors or trace tails. Chatterjee [10] provides the replacement-method context, not the critical dynamic derivative estimates.

The 2026 primary records of Du–Huang [11, 12] concern critical overlap laws and free-energy fluctuations. Their equilibrium results should not be promoted to a generator convergence theorem. Wang [13] reports optimal random-site mixing for every fixed $\beta < 1/2$, uniformly in external fields. Boban–Li–Oveis Gharan [14] report polynomial mixing up to $\beta \le 1/2 + \varepsilon$ for an absolute positive

ε. Those stated subcritical regimes do not supply the endpoint $\beta = 1$ functional limit. These comparisons are based on retrieved primary abstracts/records; they are not independent full-paper audits. The publisher PDF open for Kuwae–Shioya failed after its bibliographic search result was found; no detailed new assertion is based on that failed retrieval. M428 and its companion inputs are retained at the catalogue-pinned public snapshot, not silently replaced by a moving main-branch source.

10.2 What the Ram formulation usefully adds

- It makes the physical measure and metric auditable. Moment coordinates in (5.3) and localized kernels in (5.5) have exact mass and density factors.
- It separates observable-index tails from trial approximation. Theorem 6.2 controls one; Theorem 7.1 controls the other. This distinction is essential when the raw source vector has infinite mass.
- It gives lower bounds that survive nonorthogonal representation and a route to upper bounds through full dual residuals and relative matrix errors. Parity reduces the actual trial banks.
- It retains the exact clock. A numerical fit cannot absorb a missing factor of n, or replace c_* by unrelaxed energy, without changing the statement being tested.

10.3 Costs, risks, and concrete next work

The infinite conditioned density and its derivatives are not inexpensive inputs; naive product quadrature solves a different problem. Gram matrices can be poorly conditioned, and a basis change may improve or worsen conditioning. Neither sparse coefficient rows nor an exact finite inverse establishes a stable infinite frame. The rigorous tail constant here is realization dependent and currently not evaluated with interval bounds. Our proof also relies on the source's closed-domain and equilibrium construction; it is not an independent verification of those long arguments.

The most useful next tasks are specific:

1. Validate finite-marginal conditioning densities and inverse moments with explicit remainder bounds for the remote square-sum distribution. Use them to assemble interval bounds for G, K, and b in a small parity-adapted cylinder bank.
2. Construct a computable full V_z' residual majorant. Enlarged-bank residuals alone are insufficient; supply an analytic complementary-space bound and combine it with (6.2).
3. Audit the common-event quantifiers in the finite cap and microscopic arguments, particularly fixed degree versus increasing degree, and the passage to disorder-dependent optimizers. Then validate any claimed finite-n error rate separately.
4. Obtain justified upper and lower information about c_* from its determinate moment problem and corrector construction. Finitely many moments alone need not identify an atom at zero.
5. Compare computational cost against the same Galerkin space in whitened conventional coordinates. Any claimed speed or conditioning gain must hold at equal physical accuracy, not after changing the model or norm.

The honest conclusion is a useful, exact representation and certificate framework for the trace step, with a proved conditional limiting tail bound. It is not a shortened independent proof of functional universality.

11 Provenance and Reproduction

11.1 Read scope and immutability

The public read scope is all of `main.tex`, `preamble.tex`, `references.bib`, `sections/01-introduction.tex` through `sections/10-support.tex`, and `figures/spectral-cap.tex`, inside the assigned preprint build. The private read scope is both indicated Volume 21 chapters in full and TR8 lines 53–235, 425–559, 784–830, and 864–872. These cover the actual elliptic/domain/Green/heat, compatible-coordinate, and residual passages used above. Sources named inside those private passages were not all opened and are not falsely listed as read. Companion CM/CS proofs were not independently audited.

The parent-supplied `M428-parent-euler-bound.txt`, `M428-parent-check.py`, and `M428-parent-check.json` were read in full from `_audit/`. They were neither modified nor rerun. Their input hashes and the source of their reported test counts are preserved. The final independent parent-extension review and its supplemental checker were also read, without execution or modification. Its M428 disposition is conditional PASS for the spectral constants, bracket, and full-residual argument, retaining the uniform-in-time qualification. This does not certify the source universality theorem; unrelated M437 findings are not used here.

No original was modified. No private passage was put into a public query or upload. The only public searches were on public mathematical topics, titles, and identifiers. No shared catalogue, queue, progress file, or helper was edited; no delivery status is set by this report's author.

The audit directory `_audit/M428-author/` contains input verification, the finite script/results, URL and native-tool status records, visual QA, and a final SHA-256 manifest. The manifest records absolute input/output paths, source read scope, URLs, audit/build hashes, and owned changed paths. It is written only after final PDF checks; its own hash is reported outside the file to avoid a recursive hash. No report/audit/build file is to be modified after that freeze.

11.2 Build and tool limitations

Bundled-runtime discovery found the installed Python, Poppler, PDF skill, and artifact marker locally. The workspace-dependency/native editor tools were not exposed in this author session's discovered tool inventory. The parent then made the bounded native calls against this actual report: `open_in_codex` returned `queued`, which does not confirm visible editor display; `compile_latex_document` returned `compile-failed` with "Unable to find standard directories for platform." This is a native platform failure, not a TeX diagnostic. Native compilation is not claimed verified. No TeX installation or plugin was installed.

The separately requested on-disk PDF is exported through the unchanged `_tools/compile_report.py`, using the supplied bundled Python and installed MiKTeX. Round 1 and round 2 each use the helper's three internal TeX passes. After round 2, the parent's requested provenance corrections required one provenance-only re-export under round 2, with no further mathematical or formatting iteration; the preceding round-2 build is retained separately. Their logs, reference/overflow audits, extracted text, all-page PNGs, and contact sheets stay within `_build/TR-M428-Ram-SK/`. Visual QA covers every final page, with detailed inspection of dense mathematics and tables. The native limitation does not turn a successful MiKTeX export into a native compilation success.

References

[1] OpenAI, *Functional universality of critical SK autocorrelations*, October 5, 2026. Public source at commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned build directory. Target labels: `eq:source-40`,

`prop:trace-goe`, `lem:trace-fixed`. Source claims and dependencies, not independently certified in their entirety by this report.

[2] OpenAI, *Critical slowing down in the Sherrington–Kirkpatrick model*, September 24, 2026. Pinned companion PDF. Cited here for dependencies stated within M428, not independently read in full.

[3] OpenAI, *Critical mixing in the Sherrington–Kirkpatrick model*, September 25, 2026. Pinned companion PDF. Cited here for dependencies stated within M428, not independently read in full.

[4] M. Subbarao, private Ram Calculus books, Volume 21, *A Nonorthogonal Representation Carries a Metric* and subsequent sections, local `ch01d-physical-metrics.tex`. Full chapter read. Physical Gram, omitted residual, metric approximation, finite measurements.

[5] M. Subbarao, private Ram Calculus books, Volume 21, *A Dense Sequence With a Spurious Spectral Value* and subsequent sections, local `ch01f-spectral-limits.tex`. Full chapter read. Strong/norm resolvents, stronger domain contracts, and target-specific duality.

[6] M. Subbarao, private *Report 8: PDE-First Ram Formulations*, local `RamNonlin-IE-PDE-astra/TR8-lin-non-lin/TR8-lin-non-lin.tex`. Selected passages and exact line ranges given in Section 11 and the manifest.

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[14] M. Boban, A. Li, and S. Oveis Gharan, *Rank-1-perturbed trickledown theorems: Mixing time of Glauber dynamics for the Sherrington–Kirkpatrick model up to $\beta \le 1/2 + \varepsilon$*, 2026, v1. https://arxiv.org/abs/2609.13138v1.

[15] Parent review, *M428 parent local spectral extension*, October 7, 2026. Local read-only `_audit/M428-parent-euler-bound.txt`, with `M428-parent-check.py` and `M428-parent-check.json`. Weighted backward-Euler curve bound, full-domain residual chains, and finite evidence; not an SK simulation or full-source validation.